

SMART MAKERSPACE

PPE Computer vision project – final presentation

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Week 14 (4/17/26)



Don't need to reiterate over this every week, only when there are changes

Problem Statement (Scope for This Iteration)

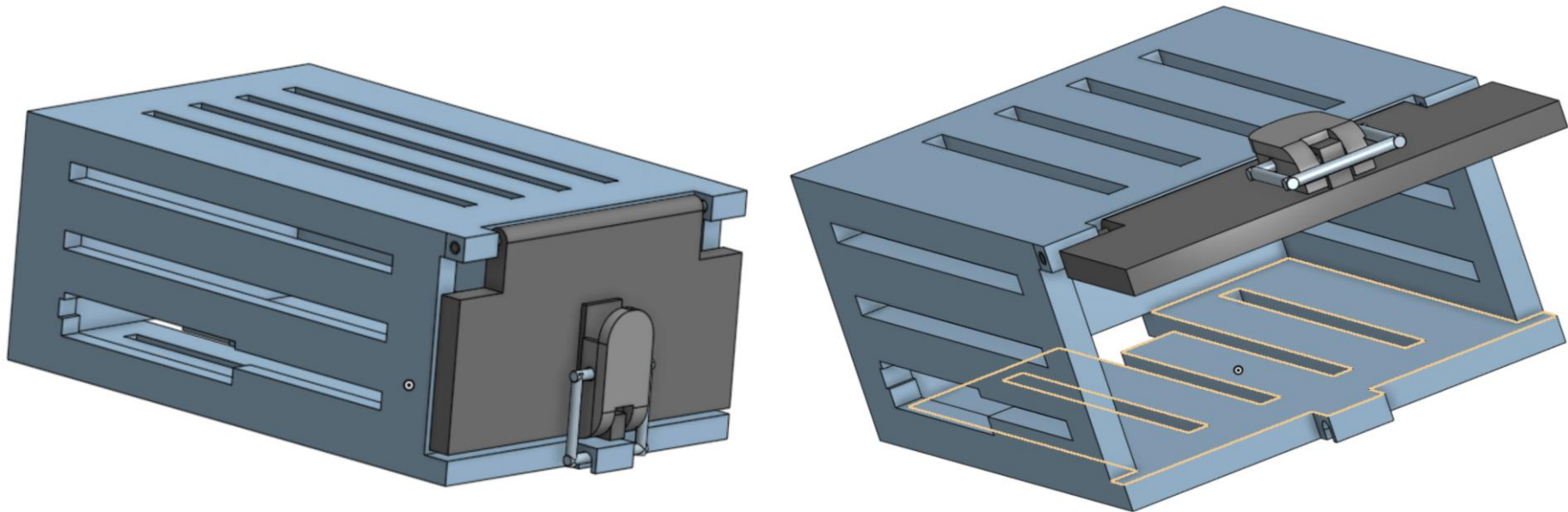
- Makerspaces have difficulty detecting whether an individual is wearing proper personal protective equipment (PPE) using a Raspberry Pi 5 due to both hardware limitations and insufficient training data. Therefore, our team will continue collecting in-shop training images and evaluate how hosting on different servers(online/local) and how different hardware platforms affect the accuracy of real-world deployment.

Projected Impact (What is beneficial about this project?)

- By automating PPE compliance monitoring, the system can reduce the need for manual checks, potentially saving **~260 staff hours and ~\$3900 per year**.
- improving real-time safety enforcement in makerspaces.

Update – pass two weeks

- Completed data to inference pipeline
- Connected pi detection to a LED display
- Working on building a new Pi 5 case



Project summary

Successes

- Improved model accuracy by relabeling and retraining on previously collected dataset
- Successfully implemented a training-to-inference pipeline on the Hailo AI Hat + 2 accelerator
- 3d desinge a new pi case

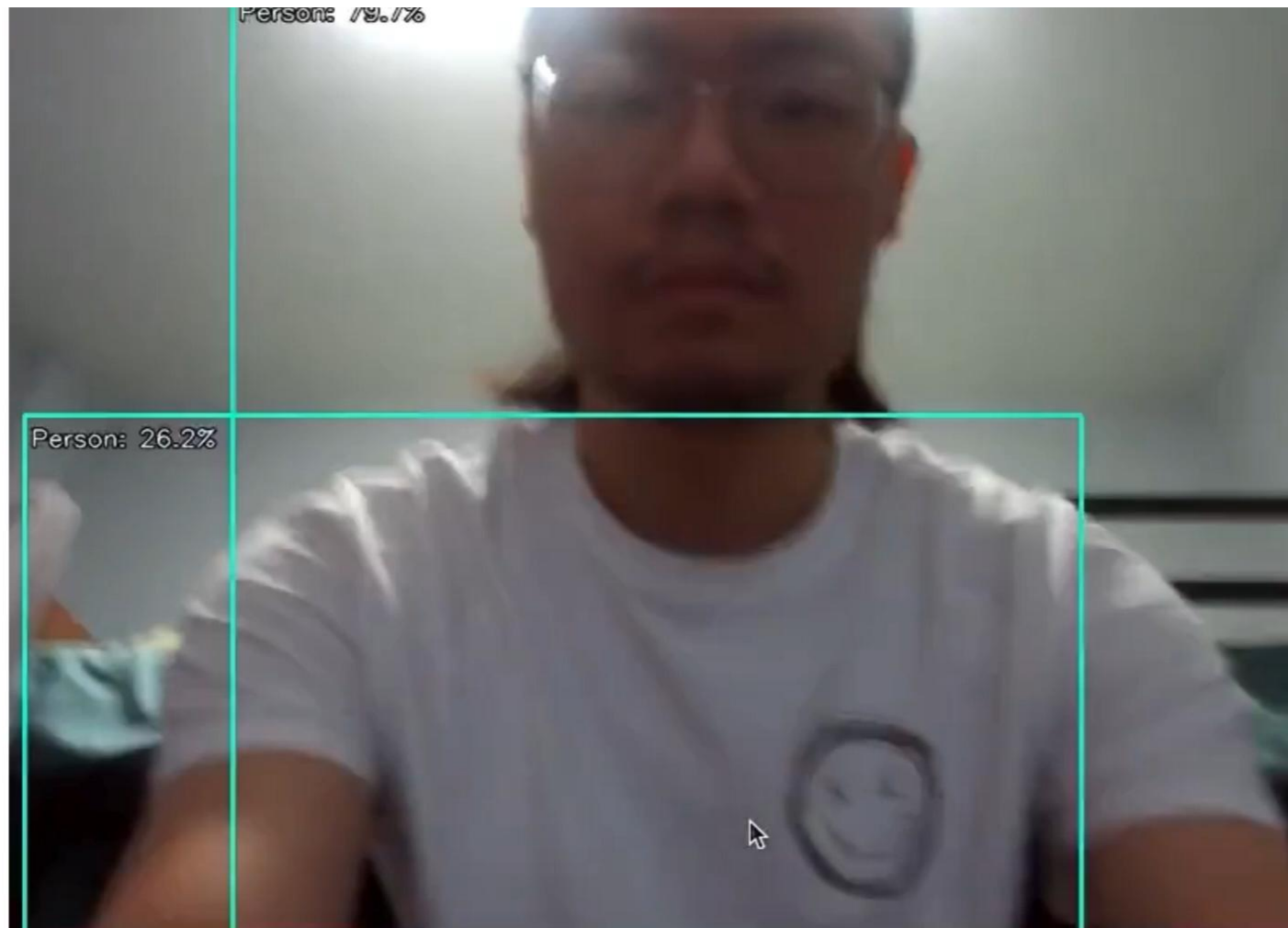
Areas for improvement

- Deploy the system in makerspace to gather feedback
- Collect higher-resolution and more diverse training data to further improve model accuracy
- Adjust camera position for more reliable detection

Plan for next semester

- Collect higher-resolution images for training
- Deploy the system in the makerspace
- Test the Jetson Nano, if finish with current setup, for a multi-camera system

New inference



Interacting with LED





Questions?